

STA301-PROBABILITY

SOLVED MCQS FROM FINAL TERMS PAPERS

SOLVED BY JUNAID MALIK AND TEAM



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- 1) SRS
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Specification)
- 2) DD
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- 3) Test phase + viva
- 4) Viva preparation
- 5) Final Deliverable

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

Q1: The marginal p.d.f of any continuous variable of obtained by

.....

- Averaging
- Integrating
- Multiplying
- Dividing

Q2: in a normal distribution how area lies between $\mu \pm \sigma$

- 65%
- 68.26%
- 75%
- 80%

Q3: A correlation coefficient

- Tells you the direction of the slope of the scattergram
- Is a sort of index how of close the point of a scatter gram deviated from straight line through those points.
- Efficiently summaries some of the in scatterplot.
- All of these

Q4: The values of moment ratios b1 and b2 of normal distribution are:

- 0 and 1
- 0 and 2
- 0 and 3
- 0 and 4

Q5: which of the following is not a property of a binomial experiment?

- The experiment consists of a sequence of an identical trials
- Each outcome can be referred to as a success or a failure

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- The probabilities of the two outcomes can change from one trial to the next.
- All of the above

Q6: we can obtain the individual probabilities of X and Y from the joint probability function / distribution of (x, y) such individual probabilities are known as

- Marginal probabilities
- Bivariate probabilities
- Separate probabilities
- Independent probabilities

Q7: Assume a large distribution in which $\sigma = 6$. What is the standard error of the mean based for which $n = 10$?

- 167
- 15
- 3
- 1.9

Q8: In a bivariate probability distribution of X and Y,

If $f(1, 1) = 4/15$, $g(1) = 2/3$, $h(1) = 2/5$

Then

- X and Y are statistically independent
- X and y are not statistically independent
- Both of these
- None of these

Q9: In hyper geometric distribution events can be classified in to

- A single category
- 2 categories
- 3 categories
- 4 categories

Q10: In binomial distribution, formula of calculating mean is:

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➤ $\mu = p + q$

➤ $\mu = np$

➤ $\mu = p q$

➤ $\mu = q n$

Q11: which of the following is not a property of binomial experiment?

- The experiment consists of a sequence of n identical trials
- Each outcome can be referred to as a success or a failure
- The probabilities of the outcomes can change from one trial to the next
- All of the above

Q12: Poisson distribution can be used to approximate the hyper geometric distribution when , $n < 0.05$, $n > 20$ and

- $P > 0.05$
- $P < 0.05$
- $P > 0.10$
- $P < 0.10$

Q13: the hypergeometric probability distribution has parameters.

- N, n
- N, P, n
- N, n, k
- $N, N-k, n$

Q14: A random experiment is one in which we repeat it a large number of times under similar condition and it produces.

- Same result
- Different result
- Independent result
- None of the above

Q15: Given below is the probability distribution of a random variable X , what is the value of “ K ”

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X	0	1	2	3	4
P(x)	0.5	0.2	0.07	k	0.03

- 0.02
- 0.1
- 0.20
- 0.002

Q16: Poisson distribution is never:

- Symmetrical
- Positively skewed
- Negatively skewed
- Asymmetrical

Q17: if X and Y are independent, then Cov (X, y) is equal to:

- 0
- 1
- 0 and 1
- None

Q18: if $E(x) = 3$ then $E(4X + 3) = \dots\dots\dots$

- 15
- 12
- 8
- 3

Q19: Which of the following is not a requirement of a binomial distribution?

- A constant probability of success
- Only two possible outcomes
- A fixed number of trials
- Equally likely outcomes

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Q20: which of the following is the requirement of a uniform distribution?

- A constant probability of success
- Only two possible outcomes
- A fixed number of trials
- Equally likely outcomes

Q21: if we are testing $H_0 : \text{mean} = 250$ against

$H_1 : \text{mean} > 250$ (exceeds 250) . Then Rejection region will be

- In the center
- On both sides of mean
- On left side of mean
- On right side of mean

Q22: Suppose $H_0: \mu_1 < \mu_2$ $H_1 : \mu_1 > \mu_2$ and critical value is value = 1.645 if calculated value of $Z = 1.88$ then what will be your conclusion?

- Accept H_0
- Reject H_0
- Impossible to decide
- None of the above

Q23: The t-distribution can never become narrower than:

- F-distribution
- Standard normal distribution
- Chi-square distribution
- Exponential distribution

Q24: t-distribution can be used to find the confidence interval for:

- Population variance
- Population mean
- Difference between population proportions
- Populating proportion

Q25: which one the following provides the basis for hypothesis testing?

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- Null hypothesis
- Alternative hypothesis
- Critical value
- Test – statistic

Q26: for test hypothesis $H_0: \mu_1 = \mu_2$ and $H_1: \mu_1 < \mu_2$, the critical region at 0.05 level of significance and $n > 30$

- Z less and equal 1.96
- $Z > 1.96$
- $Z > 1.645$
- $Z < -1.645$

Q27: Alpha is the probability of.....

- Making type I error
- Making Type II error
- Accepting H_0 when it is true
- Rejecting H_1 when it is wrong

Q28: The test statistic to test the $\mu_1 = \mu_2$ (U represent the mean of population normal population when $n > 30$).

- Z- test
- T – test
- F – test
- All of above

Q29: Suppose that the workers of factory B believes that the average income that the average income of the workers of factory A exceed that average income the null hypothesis will be

- $\text{MeanA} > \text{MeanB}$
- $\text{MeanA} \supseteq \text{MeanB}$
- $\text{MeanA} < \text{MeanB}$
- $\text{MeanA} \subseteq \text{MeanB}$

Q30: which of the following statement is NOT true?

- The t distribution is Symmetric about Zero

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- The t distribution is more spread out than the standard normal distribution
- As the degrees of freedom get smaller , the t-distributions dispersion get smaller
- The t distribution is mound - shaped

Q31: A----- is a range of number inferred from the sample that sample that has a certain probability of including the population parameter over the long run.

- Hypothesis
- Lower limit
- Confidence interval
- Probability limit

Q32: The proportion of cigarette smokers in Pakistan is greater than 20% , the null hypothesis Ho is,

- $P < 0.20$
- $P \neq 0.20$
- $P \leq 0.20$
- $P > 0.20$

Q33: A null hypothesis is generally denotes by:

- H_0
- H_1
- H_2
- H_3

Q34: A deserving player was not selected in the cricket team .it is an example of

- Type-I error
- Type – II error
- Type – III error
- Type – Iv error

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Q35: As sample size goes up, what tends to happen to 95 %confidence interval?

- They become wider
- They become more precise
- They become more precise and narrow
- They become more narrow

Q36: to determine a sample size which test we use-----

- Z
- I
- F
- None these

Q37: How can we interpret 90% confidence interval for the mean of the normal population?

- There are 10 % chances of falling true value of the parameter
- There are 90% chances of falling true value of the parameter
- There are 100 chances of falling true value of the parameter
- All are correct

Q38: for the smaller values of v , the t distribution will be -----than the standard normal distribution.

- Less spread out
- More spread out
- Same length
- None of these

Q39: if we are testing $H_0 : \text{mean} = 50$ against

$H_1 : \text{mean} < 50$ then rejection region will be

- In the center
- On left side of mean
- On right side of mean
- On both side

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Q40: if we are testing $H_0 : p_1 < p_2$ (or) $p_1 - p_2 < 0$ against $H_1 : p_1 > p_2$ (or) $p_1 - p_2 > 0$ and the tabulated of $Z = 1.645$ and Z Calculated value = 2.63 then what will be conclusion\end {gathered}\]

➤ **Reject H_0**

- Accept H_0
- None of the above
- Impossible to decide

Q41: In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to

- 3
- **5**
- 10
- 15

Q42: if the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldier but not more than 3 inches. What will be it's ALTERNATIVE hypothesis?

- **$P > 3$**
- $P < 3$
- $P = 3$
- None of this

Q43 : the critical region is computed from the value of .

- **Alpha**
- Beta
- $1 - \text{Alpha}$
- $1 - \text{beta}$

Q44: To the test hypothesis about the difference of means for large sample , what test statistics will be used?

- Chi- square
- F
- **Z**
- T

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Q45: When we start hypothesis testing, we always assume that:

- HO is true
- HO is false
- H1 is true
- All are correct

Q46: to test hypothesis about the different of means for small samples, test statistics will be used:

- F- test
- Z- test
- T-test
- All of these

Q47: ideally, the width of confidence interval should be:

- 0
- 1
- 98
- 100

Q48: conventionally, the probability of making a type- I error is denoted by which of following Symbol?

- θ (theta)
- α (Alpha)
- β (beta)
- σ (sigma)

Q49: in a random sample of 500 men from Lahore city, 300 are found to smoker ; the proportion of smoker is equal to:

- 0.3
- 0.4
- 0.6
- 0.5

Q50: if we are testing $H_0 : P_1 - p_2 = 120$ against $H_1 : P_1 - P_2 > 120$ (exceeds 120) then rejection region:

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- In Center
- On right side $Z=0$
- On left side $Z=0$
- Both side $Z=0$

Q51: In testing $H_0: \mu = 100$ against $H_1: \mu \leq 100$ at the 10% level of significance, H_0 is rejected if:

- The p-value is greater than 0.10
- The p-value is less than 0.10
- 100 is contained in the 90% confidence interval
- The value of the test is static in the acceptance region

Q52: In the method of moments, how many equations are required for finding two unknown

- 2
- 4
- 1
- 3

Q53: To determine a sample size in estimating population mean, we use the _____ value.

- X
- Z
- F

- T

Q54: The method of maximum likelihood (ML) is used to find out:

- Random numbers

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- Sample values
- Point estimates
- Correlation coefficient

Q55: Alpha is the probability of

- Making type I error
- Accepting H_0 when it is true
- Rejecting H_1 when it is wrong
- Making type II error

Q56: It is the probability of:

- Accept H_0 H_0 is true
- Reject H_0 H_0 is true
- Accept H_0 H_0 is false
- Reject H_0 H_0 is true

Q57: In interval estimation we obtained a of values as an estimate of parameter.

- Both a and b
- Range
- Group
- None of these

Q58: We can apply method of Maximum likelihood on:

- Qualitative variables only
- Discrete variables only

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➤ Discrete as well as continuous variables

➤ Continuous variables only

Q59: Inferential statistics involves:

➤ Testing of hypothesis

➤ All of these

➤ Confidence interval

➤ Estimation

Q60: Ideally, the width of confidence interval should be

➤ 98

➤ 1

➤ 100

➤ 0

Q61: Sample proportion is a/an _____ estimator of population mean.

➤ Consistent

➤ Unbiased

➤ Unbiased and consistent estimator

➤ None of these

Q62: If $\text{var}(T_1) > \text{Var}(T_2)$, where T_1 and T_2 are two unbiased estimators, then:

➤ T_1 is more consistent

➤ T_2 is more consistent

➤ T_1 is more efficient

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- T2 is more efficient

Q63: The confidence interval become wide and less precise by

- None of these
- Increasing the sample size
- Decreasing the level of significance
- Decreasing the sample size (by increasing the sample size the confidence interval become more precise)

Q64: From which of the following methods we can obtain point estimate of the population parameters:

- All of the above
- Maximum likelihood method
- Method of moments
- Method of least squares

Q65: A confidence interval has a specified probability _____ of containing the true value of parameter.

- $1-\alpha$
- $\alpha/2$
- α
- $1+\alpha$

Q66: _____ are equivalent.

- Type-II error and level of confidence
- Type-I error and level of significance
- Type-II error and level of significance

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- Type-I error and level of confidence

Q67: In a geometric distribution, maximum likelihood estimator (MLE) for proportion (P) is equal to:

- Sample variance
- Sample mean
- Reciprocal of the sample variance
- Reciprocal of the mean

Q68: Internal estimation and Confidence interval are:

- Different
- Independent
- Dependent
- Same

Q69: “A failed student was passed by the examiner”, is an example of:

- Type-I error
- Type-IV error
- Type-III error
- Type-II error

Q70: A confidence interval has a specified probability _____ of containing the true value of parameter.

- a
- $a/2$
- $1+a$
- $1-a$

Q71: When sample size is to be considered large, population standard deviation can be replaced by

- Sample variance

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- Sample Mean
- Population variance
- Sample Standard Deviation

Q72: As sample size goes up, what tends to happen to 95% confidence intervals?

- They become more narrow
- They become more precise and narrow
- They become more precise
- They become wider

Q73: An industrial designer wants to determine the average amount of time it takes an adult to assemble an “easy to assemble” toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73 assuming normality of assembly times. What is a 95%.

- (14.1123,22.7277)
- (17.1123,24.7277)
- (17.1123,22.7277)
- (17.1123,30.7277)

Q74: Suppose that the worker of factory b believes that the average income of the worker of factory A exceed their average income, the hypothesis will be:

- Mean A > Mean B
- Mean A $\geq$ Mean B
- Mean A < Mean B
- Mean A $\leq$ Mean B

Q75: β is the probability of:

- a) Reject H_0/H_0 is true
- b) Reject H_0/H_0 is false
- c) Accept H_0/H_0 is false
- d) Accept H_0/H_0 is true

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Q76: An estimator is said to be efficient if it has:

- a) Smallest variance
- b) Unbiased variance
- c) Both (a & b)
- d) None of these

Q77: Conventionally, the probability of making a type-1 error is denoted by which of the following symbol?

- a) S(sigma)
- b) β (beta)
- c) θ (theta)
- d) α (alpha)

Q78: The end points that bound the confidence interval are called:

- a) Lower limit
- b) Bounded limit
- c) Lower and upper limits
- d) Upper limits

Q79: An estimator is said to be _____ if its expected value is equal to true value of its parameter.

- a) Efficient
- b) Consistent
- c) Biased
- d) un Biased

Q80: The width of the interval is called _____ of the estimate.

- a) Biased
- b) Accuracy
- c) Precision
- d) Unbiasedness

Q81: If we draw a sample of size “n” from a population then sample is called “large sample” it.

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- a) $n > 40$
- b) $n > 30$
- c) $n > 50$
- d) $n > 25$

Q82: The method of maximum likelihood (ML) is used to find out:

- a) point estimates
- b) random number
- c) correlation coefficient
- d) sample value

Q83: The precision of an estimates can be increased by increasing the:

- a) sample value
- b) size of population
- c) number of parameters
- d) size of sample

Q84: which of the following is most important and widely used methods in point estimates?

- a) The method of least squares
- b) The method of moments
- c) The method of maximum likelihood
- d) sample value

Q85: In a random sample of 500 men from Lahore city, 300 are found to be smokers: the proportion of smokers is equal to

- a. 0.6
- b. 0.3
- c. 0.5
- d. 0.4

Q86: An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9 and a 95% confidence of (2.48, 3.32).

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- a. None of these
- b. We are sure the mean yield of this new Variety is between 2.48 and 3.32
- c. We are 95% confidence that the true mean yield of this is 2.6
- d. We are 95% confident the true mean valid of this variety is between 2.48 and 3.32

Q87: A Is a range of number is inferred from the sample that has a certain probability of including the population parameter over the long run.

- a. Hypotheses
- b. Lower limit
- c. Confidence interval
- d. Probability limit

Q88: To determine a sample size which test we use.....

- a. t
- b. F
- c. None of these
- d. z

Q90: A statistic whose standard error decreases with an increase in the sample size will be...

- a. consistent
- b. unbiased
- c. efficient
- d. none

Q91: A deserving player was not in the cricket team . it is an example of:

- a. type I error
- b. type-I error
- c. type-iv error
- d. type-III error

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Q92: In a geometric distribution, maximum likelihood estimator for proportion (P) is equal to:

- a. sample variance
- b. reciprocal of the mean
- c. reciprocal of the mean variance
- d. sample mean

Q93: The precision of an estimator can be increased by decreasing the:

- a. level of significance
- b. number of estimators
- c. number of parameters
- d. size of sample

Q94: Conventionally, the probability of making a type-error is denoted by:

- a. beta
- b. theta
- c. alpha
- d. sigma

Q95: A statistic whose standard error decreases with an increase in the sample size will be

- a. Efficient
- b. None
- c. Unbiased
- d. consistent

Q96: Sample mean is a estimate of population mean.

- a. Biased
- b. None of these
- c. Unbiased
- d. Either unbiased or biased

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Q97: If we draw a sample of size 'n' from a population then sample is called 'large sample' if;

- a. $N > 30$
- b. $N > 501$
- c. $N > 40$
- d. $N > 25$

98: When sample size is to be considered large population standard deviation can be replaced by

- Population variance
- Sample variance
- Sample mean
- Sample standard derivation

Q99: Suppose that the worker of factory B believe that the average income of the worker of factory A exceeds their average income the hypothesis will be

- $\text{MeanA} > \text{MeanB}$
- $\text{MeanA} > / \text{MeanB}$
- $\text{MeanA} < \text{MeanB}$
- $\text{MeanA} < / \text{MeanB}$

Q100: For particular hypothesis test $\alpha = 0.09$ and $\beta = 0.03$ what is the value of type of error

- 0.09
- 0.91
- 0.03
- 0.97

Q101: Suppose $H_0: u_1 < u_2$ $H_1: u_1 \geq u_2$ and critical value is = 1.645 if calculated value of $z = 1.88$ then what will be your

- Accept H_0
- Reject H_0
- Impossible to decide
- None of the above

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Q102: Suppose $H_0: p=0$ $H_1: p>0$ and critical value is $Z=2.33$ if calculate the value of $Z=-2.60$ then what will be your calculation

- Accept H_0
- **Reject H_0**
- Impossible to decide
- None of the above

Q102: If the null hypothesis is that the the average height of Pakistan soldiers exceed the average height of American soldiers by more than 3 inches what will it's alternative hypothesis

- **$p>3$**
- $p<3$
- $P=3$
- None of the above

Q103: Pairing is done only in case of observation

- **Dependent**
- Independent
- Small
- Large

Q104: The critical region is computed from the value of

- **alpha**
- Beta
- $1-\text{Alpha}$
- $2-\text{Beta}$

Q105: When examining group difference where not direction of the difference is specified which of the following is used

- one tail test
- **Two tail test**
- Directional hypothesis
- Difference of mode test

Q106: If we traveling at $H_0: \text{mean} = 50$ against

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H1: $\text{mean} < 50$ then the rejection region will be

- In the center
- On left side of mean
- On right side of Mean
- On both sides

Q107: If our significance level of 5% is used rather than 1% the null hypothesis is

- Less likely to be rejected
- Just as likely to be rejected
- More likely to be rejected
- None of these

Q108: How can we interrupt our 90% confidence interval for the mean of the normal population

- There is 10% chance of falling true value of the parameter
- There are 90% Johns are falling true value of the perimeter
- There are 100 chance of finding true value of parameter
- All are correct

Q109: When examining group difference where no direction of the difference is specified which of the following is used

- One tailed test
- Two tailed test
- Directional hypothesis
- Difference of mode test

Q110: T distribution can never become narrower then

- F distribution
- Standard normal distribution
- Chi-Square distribution
- Exponential distribution

Q111: The term $1 - \beta$ is called

- Level of the test

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- Power of the test
- Size of the test
- Critical region

Q112: The T test for independent sample assume which of the following

- Score are independent
- Scorer dependent
- Groups have equal medians
- Alpha is the probability of

Q113: The level of significance is also called

- Power of test
- Type II error
- Type I letter
- Acceptance region

Q114: For a sample $p_1=0.2, n_1=100$ and for second sample $p_2=0.5$ and $n_2=200$ the pooled estimator p_c is

- 0.30
- 0.45
- 0.40
- 0.35

Q115: The location of the critical region depends upon

- Null hypothesis
- Alternative hypothesis
- Value of alpha

Q116: Meaning of T distribution will not exist when D.F will equal to

- 1
- 2
- 3
- 4

Q117: Which of the following is not a criterion for a good problem situation?

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- It must be authentic
- It must be clearly structured and defined
- It must be clearly structured and defined
- It must be appropriate to student of development

Q118: To test a hypothesis about the difference of mean for small samples with test statistic sticks will be used

- z- test
- t-test
- F-test
- All of these

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